SPRINT 2: MODEL BUILDING & EVALUATION

Hotel Booking Cancellation Prediction System

Dataset: hotel_bookings.csv | Target: is_canceled (1 = canceled, 0 = not canceled)

1. Goal

Build a baseline classifier and compare multiple classification algorithms to identify a suitable model for predicting hotel booking cancellations.

2. Dataset and preparation

The supplied CSV contains 119,390 rows and 32 columns. The Sprint 1 cleaning workflow removed 31,994 exact duplicate rows, leaving 87,396 rows. The 80:20 stratified split produced 69,916 training rows and 17,480 testing rows. The target is binary. The same split and target distribution were used for every model.

3. Baseline model

Logistic Regression was used as the baseline classification model. It achieved 78.04% test accuracy and 67.42% F1-score. The baseline provides a reference point for the other algorithms.

4. Models trained

The following classification models were trained: Logistic Regression, Decision Tree, Random Forest, Gradient Boosting, Linear SVM and Gaussian Naive Bayes. Linear Regression is not included because the project predicts a binary class, not a continuous numeric value.

5. Evaluation metrics

Accuracy measures the overall proportion of correct predictions. Precision measures how many bookings predicted as canceled were actually canceled. Recall measures how many actual cancellations were detected. F1-score balances precision and recall. The confusion matrix reports true negatives, false positives, false negatives and true positives. ROC-AUC measures ranking ability across classification thresholds.

6. Model comparison

Model	Train Acc.	Test Acc.	Precision	Recall	F1	ROC-AUC	Acc. gap
Random Forest	0.865	0.821	0.635	0.819	0.716	0.905	0.044
Decision Tree	0.796	0.781	0.567	0.858	0.683	0.886	0.015
Logistic Regression	0.782	0.780	0.569	0.827	0.674	0.870	0.002
SVM (Linear)	0.779	0.775	0.562	0.829	0.670	0.868	0.004
Gradient Boosting	0.828	0.827	0.765	0.536	0.630	0.877	0.000
Naive Bayes	0.487	0.484	0.339	0.922	0.496	0.742	0.003

Accuracy gap = training accuracy – testing accuracy. A large positive gap can indicate overfitting; a very low score on both sets can indicate underfitting. This is a diagnostic, not a definitive test.

7. Overfitting and underfitting analysis

Random Forest: train accuracy 0.865, test accuracy 0.821, gap 0.044. Some overfitting; monitor generalization.

Decision Tree: train accuracy 0.796, test accuracy 0.781, gap 0.015. No strong overfitting signal from accuracy gap.

Logistic Regression: train accuracy 0.782, test accuracy 0.780, gap 0.002. No strong overfitting signal from accuracy gap.

SVM (Linear): train accuracy 0.779, test accuracy 0.775, gap 0.004. No strong overfitting signal from accuracy gap.

Gradient Boosting: train accuracy 0.828, test accuracy 0.827, gap 0.000. No strong overfitting signal from accuracy gap.

Naive Bayes: train accuracy 0.487, test accuracy 0.484, gap 0.003. Likely underfitting / weak model fit.

8. Confusion matrices

Matrix layout is [[TN, FP], [FN, TP]] for the positive class 'canceled' (1).

Random Forest: [[10415, 2260], [868, 3937]]

Decision Tree: [[9528, 3147], [681, 4124]]

Logistic Regression: [[9670, 3005], [833, 3972]]

SVM (Linear): [[9569, 3106], [820, 3985]]

Gradient Boosting: [[11886, 789], [2230, 2575]]

Naive Bayes: [[4036, 8639], [377, 4428]]

9. Best model and conclusion

Random Forest is the selected model when F1-score is the primary selection criterion. It achieved 82.11% test accuracy, 63.53% precision, 81.94% recall, 71.57% F1-score and 0.905 ROC-AUC. Gradient Boosting achieved the highest test accuracy (82.73%), but its recall/F1 trade-off is different. The final choice should reflect whether the hotel values detecting more cancellations (recall) or avoiding false cancellation alerts (precision).

10. Important implementation notes

Preprocessing is fitted inside each model pipeline on training data only, then applied to the test set. reservation_status, reservation_status_date and assigned_room_type are excluded to reduce target leakage. One-hot encoding is used for Logistic Regression and SVM; ordinal encoding is used for tree models and Gaussian Naive Bayes. Ordinal encoding imposes arbitrary category numbers, so Naive Bayes results should be interpreted cautiously. The current models are baseline comparisons, not deployment-ready systems.

How to run

```
pip install pandas numpy scikit-learn joblib
python sprint2_model_comparison.py --data hotel_bookings.csv
```

Outputs: sprint2_output/model_comparison.csv, model_metrics.csv, best_hotel_cancellation_model.joblib and best_model.txt.